training

October 12, 2026

[89]:
```
# imports
import lightgbm as lgb
import pandas as pd
import numpy as np
from sklearn.metrics import accuracy_score
from sklearn.model_selection import TimeSeriesSplit
from datetime import date, timedelta
import yfinance as yf
import shap
import matplotlib.pyplot as plt
import streamlit as st
import yfinance as yf
import plotly.graph_objects as go
from scipy import stats
import statsmodels.api as sm
```

0.0.1 Training

[90]:
```
def make_splits(df, n_splits=5, target_col='target', gap_days=21):
    """
    Create walk-forward splits from the merged DataFrame.
    gap_days: trading days between train end and test start (avoids lookahead).
    """
    # Work on unique dates to split time, not rows
    dates = df.index.get_level_values('Date').unique().sort_values()
    tss = TimeSeriesSplit(n_splits=n_splits, gap=gap_days)

    splits = []
    for train_idx, test_idx in tss.split(dates):
        train_dates = dates[train_idx]
        test_dates  = dates[test_idx]

        train_df = df[df.index.get_level_values('Date').isin(train_dates)]
        test_df  = df[df.index.get_level_values('Date').isin(test_dates)]

        splits.append((train_df, test_df))

    return splits
```

[91]:
```
ai_sector_features_signals_df = pd.read_csv('../../data/ai_sector_features_df.
 ↪csv', index_col=['Date', 'Ticker'], parse_dates=['Date'])
ai_sector_features_signals_df['regime'] = np.
 ↪where(ai_sector_features_signals_df.index.get_level_values('Date') <␣
 ↪'2023-01-01', 'pre_chatgpt', 'post_chatgpt')
splits = make_splits(ai_sector_features_signals_df, n_splits=5)
splits
```

[91]:
```
[(                    ret_1d    ret_3d    ret_5d   ret_10d   ret_20d    vol_5d
\
  Date       Ticker
  2019-01-02 NVDA    0.020374  0.038499  0.071923 -0.051261 -0.166494  0.023554
             TSM    -0.010566 -0.007069  0.034854  0.008283 -0.028465  0.020815
             AVGO   -0.003028  0.010604  0.084303  0.008072  0.079068  0.025708
             MRVL    0.019148  0.050955  0.145038  0.086345  0.028352  0.028901
             INTC    0.003196  0.015531  0.080064  0.000000 -0.045224  0.024625
  …                     …         …         …         …         …         …
  2020-02-03 NVDA    0.016495 -0.021219  0.000541 -0.035903  0.018046  0.026791
             META    0.011292 -0.085293 -0.049704 -0.080805 -0.021469  0.037305
             SMCI    0.025036 -0.004170 -0.001742 -0.000349  0.222696  0.018681
             AMD     0.021702  0.010735 -0.025173 -0.057137 -0.011934  0.040794
             GOOGL   0.034771  0.017780  0.035530  0.002082  0.088930  0.018473

                       vol_20d     rsi_14  rel_soxx   rel_qqq  …  \
  Date       Ticker                                             …
  2019-01-02 NVDA    0.034328  37.207812 -0.107088 -0.083558  …
```

```
            TSM     0.017657  49.090132  0.030941  0.054471  …
            AVGO    0.026895  58.335498  0.138474  0.162004  …
            MRVL    0.030025  64.768931  0.087758  0.111288  …
            INTC    0.025632  49.416499  0.014182  0.037711  …
…                      …         …        …         …       …
2020-02-03  NVDA    0.018825  39.591441  0.040828 -0.020240  …
            META    0.019977  30.772450  0.001313 -0.059755  …
            SMCI    0.036809  52.149698  0.245478  0.184411  …
            AMD     0.023432  48.278780  0.010848 -0.050220  …
            GOOGL   0.013527  60.290737  0.111712  0.050644  …

                    nvidia_mentions  amd_mentions  intel_mentions  \
Date       Ticker
2019-01-02 NVDA            0.621311      0.000000             0.0
           TSM             0.000000      0.000000             0.0
           AVGO            0.000000      0.000000             0.0
           MRVL            0.000000      0.000000             0.0
           INTC            0.000000      0.000000             0.0
…                             …             …              …
2020-02-03 NVDA            0.761615      0.000000             0.0
           META            0.000000      0.000000             0.0
           SMCI            0.000000      0.000000             0.0
           AMD             0.000000      2.085308             0.0
           GOOGL           0.000000      0.000000             0.0

                    gpu_mentions  capex_up_score  capex_down_score  \
Date       Ticker
2019-01-02 NVDA         0.000000             0.0               0.0
           TSM          0.000000             0.0               0.0
           AVGO         0.000000             0.0               0.0
           MRVL         0.000000             0.0               0.0
           INTC         0.000000             0.0               0.0
…                          …                …                 …
2020-02-03 NVDA         0.380807             0.0               0.0
           META         0.000000             0.0               0.0
           SMCI         0.000000             0.0               0.0
           AMD          0.000000             0.0               0.0
           GOOGL        0.000000             0.0               0.0

                    competitor_mentions  capex_net  ai_sentence_ratio  \
Date       Ticker
2019-01-02 NVDA                     0.0        0.0           0.000000
           TSM                      0.0        0.0           0.000000
           AVGO                     0.0        0.0           0.000000
           MRVL                     0.0        0.0           0.000000
           INTC                     0.0        0.0           0.003636
…                                   …          …                …
```

```
2020-02-03 NVDA                    0.0        0.0           0.000000
           META                    0.0        0.0           0.000000
           SMCI                    0.0        0.0           0.000000
           AMD                     0.0        0.0           0.000000
           GOOGL                   0.0        0.0           0.014085

                           regime
Date       Ticker
2019-01-02 NVDA    pre_chatgpt
           TSM     pre_chatgpt
           AVGO    pre_chatgpt
           MRVL    pre_chatgpt
           INTC    pre_chatgpt
…                        …
2020-02-03 NVDA    pre_chatgpt
           META    pre_chatgpt
           SMCI    pre_chatgpt
           AMD     pre_chatgpt
           GOOGL   pre_chatgpt

[4110 rows x 27 columns],
                      ret_1d    ret_3d    ret_5d   ret_10d   ret_20d    vol_5d
\
Date       Ticker
2020-03-05 MRVL    0.103586  0.135246  0.213139 -0.007959 -0.035217  0.048648
           NVDA   -0.039436 -0.011359  0.081908 -0.114177  0.090499  0.054314
           TSM    -0.023651 -0.004466  0.045787 -0.013279 -0.037811  0.030821
           DELL   -0.058322 -0.036329 -0.062213 -0.228517 -0.195708  0.054187
           AMD    -0.039912  0.013696  0.093161 -0.159944 -0.034711  0.045356
…                       …       …       …        …        …       …
2021-05-03 SMCI    0.006483 -0.023073 -0.036213 -0.033714 -0.068500  0.013098
           ASML    0.012689 -0.013551 -0.021055  0.041602  0.024721  0.013221
           TSM    -0.006853 -0.031088 -0.045918  0.004679 -0.068755  0.007179
           GOOGL  -0.004427 -0.006765  0.014351  0.023286  0.055936  0.018696
           ANET   -0.001460 -0.018862 -0.019809 -0.000318  0.009916  0.011649

                       vol_20d     rsi_14  rel_soxx   rel_qqq  …  \
Date       Ticker                                              …
2020-03-05 MRVL    0.044215  50.299788  0.054448  0.037652  …
           NVDA    0.044603  52.103749  0.180163  0.163367  …
           TSM     0.024705  41.317793  0.051854  0.035058  …
           DELL    0.034692  23.078295 -0.106043 -0.122839  …
           AMD     0.042370  40.262906  0.054954  0.038158  …
…                       …         …       …        …   …
2021-05-03 SMCI    0.012183  31.283325  0.000203 -0.084026  …
           ASML    0.021143  59.698621  0.093424  0.009195  …
           TSM     0.017407  39.275956 -0.000052 -0.084282  …
```

```
           GOOGL   0.012081  63.834998  0.124639  0.040410  …
           ANET    0.011626  50.786318  0.078619 -0.005610  …

                   nvidia_mentions  amd_mentions  intel_mentions  \
Date       Ticker
2020-03-05 MRVL            0.00000      0.000000             0.0
           NVDA            0.91659      0.000000             0.0
           TSM             0.00000      0.000000             0.0
           DELL            0.00000      0.000000             0.0
           AMD             0.00000      2.085308             0.0
…                            …            …              …
2021-05-03 SMCI            0.00000      0.000000             0.0
           ASML            0.00000      0.000000             0.0
           TSM             0.00000      0.000000             0.0
           GOOGL           0.00000      0.000000             0.0
           ANET            0.00000      0.000000             0.0

                   gpu_mentions  capex_up_score  capex_down_score  \
Date       Ticker
2020-03-05 MRVL         0.00000        0.000000               0.0
           NVDA         0.30553        0.000000               0.0
           TSM          0.00000        0.000000               0.0
           DELL         0.00000        0.000000               0.0
           AMD          0.00000        0.000000               0.0
…                         …              …                …
2021-05-03 SMCI         0.00000        0.350263               0.0
           ASML         0.00000        0.000000               0.0
           TSM          0.00000        0.000000               0.0
           GOOGL        0.00000        0.000000               0.0
           ANET         0.00000        0.000000               0.0

                   competitor_mentions  capex_net  ai_sentence_ratio  \
Date       Ticker
2020-03-05 MRVL                    0.0   0.000000           0.000000
           NVDA                    0.0   0.000000           0.000000
           TSM                     0.0   0.000000           0.000000
           DELL                    0.0   0.000000           0.000000
           AMD                     0.0   0.000000           0.000000
…                                  …         …                …
2021-05-03 SMCI                    0.0   0.350263           0.008000
           ASML                    0.0   0.000000           0.000000
           TSM                     0.0   0.000000           0.000000
           GOOGL                   0.0   0.000000           0.000000
           ANET                    0.0   0.000000           0.007353

                        regime
Date       Ticker
```

```
  2020-03-05 MRVL    pre_chatgpt
             NVDA    pre_chatgpt
             TSM     pre_chatgpt
             DELL    pre_chatgpt
             AMD     pre_chatgpt
  …                        …
  2021-05-03 SMCI    pre_chatgpt
             ASML    pre_chatgpt
             TSM     pre_chatgpt
             GOOGL   pre_chatgpt
             ANET    pre_chatgpt

  [4395 rows x 27 columns]),
 (                     ret_1d    ret_3d    ret_5d   ret_10d   ret_20d    vol_5d
\
  Date       Ticker
  2019-01-02 NVDA    0.020374  0.038499  0.071923 -0.051261 -0.166494  0.023554
             TSM    -0.010566 -0.007069  0.034854  0.008283 -0.028465  0.020815
             AVGO   -0.003028  0.010604  0.084303  0.008072  0.079068  0.025708
             MRVL    0.019148  0.050955  0.145038  0.086345  0.028352  0.028901
             INTC    0.003196  0.015531  0.080064  0.000000 -0.045224  0.024625
  …                     …         …         …         …         …         …
  2021-04-01 SMCI    0.013825  0.056845  0.068537  0.035836  0.213979  0.019634
             AMZN    0.021628  0.027724  0.037666  0.043927  0.061604  0.010703
             MRVL    0.013883  0.067727  0.109721  0.090230  0.238404  0.022475
             INTC    0.008594  0.000775  0.040793  0.012867  0.106635  0.022477
             NVDA    0.034724  0.066689  0.101833  0.085616  0.116915  0.018220

                      vol_20d     rsi_14  rel_soxx   rel_qqq  …  \
  Date       Ticker                                           …
  2019-01-02 NVDA    0.034328  37.207812 -0.107088 -0.083558  …
             TSM     0.017657  49.090132  0.030941  0.054471  …
             AVGO    0.026895  58.335498  0.138474  0.162004  …
             MRVL    0.030025  64.768931  0.087758  0.111288  …
             INTC    0.025632  49.416499  0.014182  0.037711  …
  …                     …         …         …         …  …
  2021-04-01 SMCI    0.019241  69.156969  0.070239  0.145317  …
             AMZN    0.016149  57.174662 -0.082136 -0.007058  …
             MRVL    0.037563  58.785300  0.094664  0.169742  …
             INTC    0.023404  56.140363 -0.037106  0.037973  …
             NVDA    0.033025  63.517494 -0.026825  0.048253  …

                     nvidia_mentions  amd_mentions  intel_mentions  \
  Date       Ticker
  2019-01-02 NVDA            0.621311           0.0        0.000000
             TSM             0.000000           0.0        0.000000
             AVGO            0.000000           0.0        0.000000
```

```
           MRVL           0.000000           0.0        0.000000
           INTC           0.000000           0.0        0.000000
…                            …             …             …
2021-04-01 SMCI           0.000000           0.0        0.000000
           AMZN           0.000000           0.0        0.000000
           MRVL           0.000000           0.0        0.000000
           INTC           0.000000           0.0        0.336984
           NVDA           0.558659           0.0        0.000000

                    gpu_mentions  capex_up_score  capex_down_score  \
Date       Ticker
2019-01-02 NVDA              0.0        0.000000               0.0
           TSM               0.0        0.000000               0.0
           AVGO              0.0        0.000000               0.0
           MRVL              0.0        0.000000               0.0
           INTC              0.0        0.000000               0.0
…                            …             …             …
2021-04-01 SMCI              0.0        0.350263               0.0
           AMZN              0.0        0.000000               0.0
           MRVL              0.0        0.000000               0.0
           INTC              0.0        0.000000               0.0
           NVDA              0.0        0.000000               0.0

                    competitor_mentions  capex_net  ai_sentence_ratio  \
Date       Ticker
2019-01-02 NVDA                0.000000   0.000000           0.000000
           TSM                 0.000000   0.000000           0.000000
           AVGO                0.000000   0.000000           0.000000
           MRVL                0.000000   0.000000           0.000000
           INTC                0.000000   0.000000           0.003636
…                                 …          …                …
2021-04-01 SMCI                0.000000   0.350263           0.008000
           AMZN                0.582547   0.000000           0.000000
           MRVL                0.000000   0.000000           0.000000
           INTC                0.168492   0.000000           0.000000
           NVDA                0.000000   0.000000           0.007812

                         regime
Date       Ticker
2019-01-02 NVDA    pre_chatgpt
           TSM     pre_chatgpt
           AVGO    pre_chatgpt
           MRVL    pre_chatgpt
           INTC    pre_chatgpt
…                        …
2021-04-01 SMCI    pre_chatgpt
           AMZN    pre_chatgpt
```

```
            MRVL    pre_chatgpt
            INTC    pre_chatgpt
            NVDA    pre_chatgpt

  [8505 rows x 27 columns],
                        ret_1d    ret_3d    ret_5d   ret_10d   ret_20d    vol_5d
\
  Date       Ticker
  2021-05-04 DELL   -0.012886 -0.034800 -0.041605 -0.048876  0.078677  0.008929
             META   -0.013082 -0.033838  0.048720  0.051908  0.039509  0.036618
             INTC   -0.006287 -0.023679 -0.018458 -0.092504 -0.132093  0.009040
             GOOGL  -0.015471 -0.035913  0.006918  0.012207  0.044164  0.020020
             NVDA   -0.032723 -0.063525 -0.066995 -0.054050  0.035332  0.013645
  …                     …        …        …         …         …         …
  2022-06-30 TSM    -0.022714 -0.047869 -0.028174 -0.077970 -0.132659  0.017313
             PLTR   -0.017335 -0.057173 -0.041226  0.116995  0.072104  0.050556
             AVGO   -0.009380 -0.039350 -0.021688 -0.070005 -0.146059  0.016898
             MSFT   -0.013179 -0.030428 -0.007842  0.020138 -0.057228  0.025820
             NVDA   -0.024643 -0.101369 -0.065701 -0.082774 -0.172369  0.040663

                       vol_20d     rsi_14  rel_soxx   rel_qqq  …  \
  Date       Ticker                                            …
  2021-05-04 DELL    0.019855  60.068139  0.152735  0.080733  …
             META    0.020343  63.422139  0.113567  0.041565  …
             INTC    0.018313  20.289305 -0.058035 -0.130037  …
             GOOGL   0.012668  59.716760  0.118222  0.046220  …
             NVDA    0.027102  39.819133  0.109389  0.037387  …
  …                     …         …        …         …  …
  2022-06-30 TSM     0.020990  32.725671  0.032446 -0.050367  …
             PLTR    0.051418  54.775989  0.237209  0.154396  …
             AVGO    0.020607  25.730491  0.019046 -0.063767  …
             MSFT    0.023285  45.943664  0.107877  0.025064  …
             NVDA    0.040598  34.023752 -0.007264 -0.090077  …

                      nvidia_mentions  amd_mentions  intel_mentions  \
  Date       Ticker
  2021-05-04 DELL             0.000000           0.0             0.0
             META             0.000000           0.0             0.0
             INTC             0.000000           0.0             0.0
             GOOGL            0.000000           0.0             0.0
             NVDA             0.558659           0.0             0.0
  …                             …             …              …
  2022-06-30 TSM              0.000000           0.0             0.0
             PLTR             0.000000           0.0             0.0
             AVGO             0.000000           0.0             0.0
             MSFT             0.000000           0.0             0.0
             NVDA             2.842524           0.0             0.0
```

```
                    gpu_mentions  capex_up_score  capex_down_score  \
Date       Ticker
2021-05-04 DELL         0.000000        0.000000               0.0
           META         0.000000        0.389712               0.0
           INTC         0.171116        0.000000               0.0
           GOOGL        0.000000        0.000000               0.0
           NVDA         0.000000        0.000000               0.0
…                            …               …                 …
2022-06-30 TSM          0.000000        0.000000               0.0
           PLTR         0.000000        0.000000               0.0
           AVGO         0.000000        0.000000               0.0
           MSFT         0.000000        0.000000               0.0
           NVDA         0.284252        0.000000               0.0

                    competitor_mentions  capex_net  ai_sentence_ratio  \
Date       Ticker
2021-05-04 DELL                     0.0   0.000000           0.000000
           META                     0.0   0.389712           0.000000
           INTC                     0.0   0.000000           0.000000
           GOOGL                    0.0   0.000000           0.000000
           NVDA                     0.0   0.000000           0.007812
…                                   …          …                 …
2022-06-30 TSM                      0.0   0.000000           0.000000
           PLTR                     0.0   0.000000           0.000000
           AVGO                     0.0   0.000000           0.000000
           MSFT                     0.0   0.000000           0.014925
           NVDA                     0.0   0.000000           0.008547

                          regime
Date       Ticker
2021-05-04 DELL    pre_chatgpt
           META    pre_chatgpt
           INTC    pre_chatgpt
           GOOGL   pre_chatgpt
           NVDA    pre_chatgpt
…                        …
2022-06-30 TSM     pre_chatgpt
           PLTR    pre_chatgpt
           AVGO    pre_chatgpt
           MSFT    pre_chatgpt
           NVDA    pre_chatgpt

[4726 rows x 27 columns]),
 (                     ret_1d    ret_3d    ret_5d   ret_10d   ret_20d    vol_5d
\
Date       Ticker
```

```
2019-01-02 NVDA    0.020374  0.038499  0.071923 -0.051261 -0.166494  0.023554
           TSM    -0.010566 -0.007069  0.034854  0.008283 -0.028465  0.020815
           AVGO   -0.003028  0.010604  0.084303  0.008072  0.079068  0.025708
           MRVL    0.019148  0.050955  0.145038  0.086345  0.028352  0.028901
           INTC    0.003196  0.015531  0.080064  0.000000 -0.045224  0.024625
…                     …         …         …         …         …         …
2022-05-31 DELL    0.007261  0.153349  0.211548  0.109039  0.051369  0.055380
           NVDA   -0.007389  0.099970  0.104983  0.081557 -0.044079  0.044540
           AMD    -0.003912  0.099406  0.071421  0.080857  0.133793  0.040359
           MSFT   -0.005014  0.035616  0.043046  0.042077 -0.042067  0.013499
           TSM     0.016317  0.054087  0.041530  0.054087  0.016425  0.023288

                     vol_20d     rsi_14  rel_soxx   rel_qqq  …  \
Date       Ticker                                            …
2019-01-02 NVDA    0.034328  37.207812 -0.107088 -0.083558  …
           TSM     0.017657  49.090132  0.030941  0.054471  …
           AVGO    0.026895  58.335498  0.138474  0.162004  …
           MRVL    0.030025  64.768931  0.087758  0.111288  …
           INTC    0.025632  49.416499  0.014182  0.037711  …
…                     …         …         …         …  …
2022-05-31 DELL    0.042988  60.862025  0.022443  0.083396  …
           NVDA    0.050591  56.486139 -0.073005 -0.012052  …
           AMD     0.050986  65.682127  0.104868  0.165821  …
           MSFT    0.024282  52.983496 -0.070993 -0.010040  …
           TSM     0.023954  65.389550 -0.012501  0.048452  …

                    nvidia_mentions  amd_mentions  intel_mentions  \
Date       Ticker
2019-01-02 NVDA            0.621311      0.000000             0.0
           TSM             0.000000      0.000000             0.0
           AVGO            0.000000      0.000000             0.0
           MRVL            0.000000      0.000000             0.0
           INTC            0.000000      0.000000             0.0
…                             …             …              …
2022-05-31 DELL            0.000000      0.000000             0.0
           NVDA            2.842524      0.000000             0.0
           AMD             0.000000      0.395257             0.0
           MSFT            0.000000      0.000000             0.0
           TSM             0.000000      0.000000             0.0

                    gpu_mentions  capex_up_score  capex_down_score  \
Date       Ticker
2019-01-02 NVDA         0.000000             0.0               0.0
           TSM          0.000000             0.0               0.0
           AVGO         0.000000             0.0               0.0
           MRVL         0.000000             0.0               0.0
           INTC         0.000000             0.0               0.0
```

```
…                              …              …                 …
2022-05-31 DELL        0.000000             0.0               0.0
           NVDA        0.284252             0.0               0.0
           AMD         0.000000             0.0               0.0
           MSFT        0.000000             0.0               0.0
           TSM         0.000000             0.0               0.0

                    competitor_mentions  capex_net  ai_sentence_ratio  \
Date       Ticker
2019-01-02 NVDA                     0.0        0.0           0.000000
           TSM                      0.0        0.0           0.000000
           AVGO                     0.0        0.0           0.000000
           MRVL                     0.0        0.0           0.000000
           INTC                     0.0        0.0           0.003636
…                                   …          …                 …
2022-05-31 DELL                     0.0        0.0           0.000000
           NVDA                     0.0        0.0           0.008547
           AMD                      0.0        0.0           0.000000
           MSFT                     0.0        0.0           0.014925
           TSM                      0.0        0.0           0.000000

                          regime
Date       Ticker
2019-01-02 NVDA    pre_chatgpt
           TSM     pre_chatgpt
           AVGO    pre_chatgpt
           MRVL    pre_chatgpt
           INTC    pre_chatgpt
…                        …
2022-05-31 DELL    pre_chatgpt
           NVDA    pre_chatgpt
           AMD     pre_chatgpt
           MSFT    pre_chatgpt
           TSM     pre_chatgpt

[13189 rows x 27 columns],
                      ret_1d    ret_3d    ret_5d   ret_10d   ret_20d    vol_5d
\
Date       Ticker
2022-07-01 AMZN     0.031541  0.020112 -0.059248  0.056917 -0.127088  0.033720
           HPE     -0.022624 -0.057455 -0.078236 -0.029213 -0.126550  0.014388
           ASML    -0.054741 -0.082747 -0.126985 -0.039133 -0.225152  0.018084
           DELL    -0.075092 -0.126686 -0.149453 -0.068642 -0.166537  0.027181
           AMD     -0.036616 -0.088017 -0.153996 -0.102133 -0.321577  0.019798
…                        …         …         …         …         …         …
2023-08-30 ANET     0.008897  0.034273 -0.014381  0.031593  0.055393  0.030342
           MRVL     0.021739  0.071776 -0.067642 -0.000523 -0.076799  0.049939
```

```
           AMZN    0.001186  0.013583 -0.003321  0.000000  0.053506  0.016053
           AMD     0.006326  0.042445 -0.025953 -0.005598 -0.025240  0.038296
           GOOGL   0.009735  0.046197  0.026517  0.055789  0.058420  0.016945

                    vol_20d     rsi_14  rel_soxx   rel_qqq  …  \
Date       Ticker                                           …
2022-07-01 AMZN    0.033639  50.847139  0.094668 -0.026231  …
           HPE     0.023390  35.406910  0.095207 -0.025692  …
           ASML    0.034085  33.951146 -0.003396 -0.124295  …
           DELL    0.028062  32.126363  0.055220 -0.065679  …
           AMD     0.034312  23.969020 -0.099820 -0.220719  …
…                     …         …        …        …   …
2023-08-30 ANET    0.017329  64.531398  0.070883  0.048795  …
           MRVL    0.031995  47.706230 -0.061308 -0.083396  …
           AMZN    0.022883  41.722717  0.068997  0.046909  …
           AMD     0.029290  46.088533 -0.009749 -0.031837  …
           GOOGL   0.013750  64.360328  0.073911  0.051823  …

                   nvidia_mentions  amd_mentions  intel_mentions  \
Date       Ticker
2022-07-01 AMZN           0.000000      0.000000             0.0
           HPE            0.000000      0.000000             0.0
           ASML           0.000000      0.000000             0.0
           DELL           0.000000      0.000000             0.0
           AMD            0.000000      0.395257             0.0
…                            …             …              …
2023-08-30 ANET           0.000000      0.000000             0.0
           MRVL           0.000000      0.000000             0.0
           AMZN           0.133815      0.000000             0.0
           AMD            0.000000      1.229256             0.0
           GOOGL          0.000000      0.000000             0.0

                   gpu_mentions  capex_up_score  capex_down_score  \
Date       Ticker
2022-07-01 AMZN             0.0             0.0               0.0
           HPE              0.0             0.0               0.0
           ASML             0.0             0.0               0.0
           DELL             0.0             0.0               0.0
           AMD              0.0             0.0               0.0
…                           …               …                 …
2023-08-30 ANET             0.0             0.0               0.0
           MRVL             0.0             0.0               0.0
           AMZN             0.0             0.0               0.0
           AMD              0.0             0.0               0.0
           GOOGL            0.0             0.0               0.0

                   competitor_mentions  capex_net  ai_sentence_ratio  \
```

```
  Date       Ticker
  2022-07-01 AMZN              0.000000        0.0           0.002882
             HPE               0.000000        0.0           0.003185
             ASML              0.000000        0.0           0.000000
             DELL              0.000000        0.0           0.000000
             AMD               0.000000        0.0           0.000000
  …                               …         …                …
  2023-08-30 ANET              0.000000        0.0           0.009434
             MRVL              0.000000        0.0           0.003367
             AMZN              0.401445        0.0           0.003003
             AMD               0.819504        0.0           0.000000
             GOOGL             0.000000        0.0           0.009615

                           regime
  Date       Ticker
  2022-07-01 AMZN     pre_chatgpt
             HPE      pre_chatgpt
             ASML     pre_chatgpt
             DELL     pre_chatgpt
             AMD      pre_chatgpt
  …                       …
  2023-08-30 ANET    post_chatgpt
             MRVL    post_chatgpt
             AMZN    post_chatgpt
             AMD     post_chatgpt
             GOOGL   post_chatgpt

  [5067 rows x 27 columns]),
 (                     ret_1d    ret_3d    ret_5d   ret_10d   ret_20d    vol_5d
\
  Date       Ticker
  2019-01-02 NVDA    0.020374  0.038499  0.071923 -0.051261 -0.166494  0.023554
             TSM    -0.010566 -0.007069  0.034854  0.008283 -0.028465  0.020815
             AVGO   -0.003028  0.010604  0.084303  0.008072  0.079068  0.025708
             MRVL    0.019148  0.050955  0.145038  0.086345  0.028352  0.028901
             INTC    0.003196  0.015531  0.080064  0.000000 -0.045224  0.024625
  …                     …         …         …         …         …
  2023-08-01 SOUN    0.025751  0.071749  0.025751 -0.352304 -0.472406  0.068468
             AVGO    0.023758  0.029624  0.002189  0.018341  0.049701  0.018367
             META    0.012900  0.035289  0.095901  0.034161  0.128278  0.027079
             MSFT    0.001250  0.016993 -0.041712 -0.064396 -0.004882  0.022907
             INTC    0.000839  0.036179  0.049853  0.037681  0.064842  0.034454

                      vol_20d     rsi_14  rel_soxx   rel_qqq  …  \
  Date       Ticker                                           …
  2019-01-02 NVDA    0.034328  37.207812 -0.107088 -0.083558  …
             TSM     0.017657  49.090132  0.030941  0.054471  …
```

```
            AVGO    0.026895  58.335498  0.138474  0.162004  …
            MRVL    0.030025  64.768931  0.087758  0.111288  …
            INTC    0.025632  49.416499  0.014182  0.037711  …
…                      …         …        …         …   …
2023-08-01  SOUN    0.076594  29.490371 -0.519048 -0.506164  …
            AVGO    0.015438  61.994165  0.003059  0.015943  …
            META    0.022841  58.403308  0.081636  0.094520  …
            MSFT    0.017923  50.034766 -0.051524 -0.038639  …
            INTC    0.024545  61.228895  0.018200  0.031085  …

                    nvidia_mentions  amd_mentions  intel_mentions  \
Date        Ticker
2019-01-02  NVDA           0.621311           0.0        0.000000
            TSM            0.000000           0.0        0.000000
            AVGO           0.000000           0.0        0.000000
            MRVL           0.000000           0.0        0.000000
            INTC           0.000000           0.0        0.000000
…                             …             …             …
2023-08-01  SOUN           0.000000           0.0        0.000000
            AVGO           0.000000           0.0        0.000000
            META           0.000000           0.0        0.000000
            MSFT           0.000000           0.0        0.000000
            INTC           0.000000           0.0        0.353107

                    gpu_mentions  capex_up_score  capex_down_score  \
Date        Ticker
2019-01-02  NVDA             0.0        0.000000               0.0
            TSM              0.0        0.000000               0.0
            AVGO             0.0        0.000000               0.0
            MRVL             0.0        0.000000               0.0
            INTC             0.0        0.000000               0.0
…                            …             …                 …
2023-08-01  SOUN             0.0        0.000000               0.0
            AVGO             0.0        0.000000               0.0
            META             0.0        0.278242               0.0
            MSFT             0.0        0.000000               0.0
            INTC             0.0        0.000000               0.0

                    competitor_mentions  capex_net  ai_sentence_ratio  \
Date        Ticker
2019-01-02  NVDA               0.000000   0.000000           0.000000
            TSM                0.000000   0.000000           0.000000
            AVGO               0.000000   0.000000           0.000000
            MRVL               0.000000   0.000000           0.000000
            INTC               0.000000   0.000000           0.003636
…                                 …          …                 …
2023-08-01  SOUN               0.000000   0.000000           0.031250
```

```
            AVGO                0.000000   0.000000           0.000000
            META                0.000000   0.278242           0.007812
            MSFT                0.000000   0.000000           0.007692
            INTC                0.353107   0.000000           0.000000

                                regime
Date       Ticker
2019-01-02 NVDA     pre_chatgpt
           TSM      pre_chatgpt
           AVGO     pre_chatgpt
           MRVL     pre_chatgpt
           INTC     pre_chatgpt
…                        …
2023-08-01 SOUN    post_chatgpt
           AVGO    post_chatgpt
           META    post_chatgpt
           MSFT    post_chatgpt
           INTC    post_chatgpt

[18235 rows x 27 columns],
                      ret_1d    ret_3d    ret_5d   ret_10d   ret_20d    vol_5d
\
Date       Ticker
2023-08-31 TSM    -0.012767 -0.004575  0.011239  0.021061 -0.014845  0.010129
           MSFT   -0.003133  0.012542  0.024346  0.034335  0.005492  0.007055
           AI     -0.013045  0.061602  0.075963  0.033655 -0.210285  0.024857
           NVDA    0.001847  0.053806  0.046477  0.138681  0.108728  0.023974
           AMZN    0.021766  0.036578  0.046799  0.030079  0.070592  0.009265
…                     …         …         …         …         …         …
2024-10-29 ANET    0.023421  0.014834  0.012634  0.023604  0.051643  0.012338
           MSFT    0.012565  0.016999  0.010386  0.031547  0.026766  0.008070
           DELL    0.008039  0.008791  0.010216 -0.029943  0.078670  0.013123
           GOOGL   0.017754  0.042773  0.027492  0.025504  0.016109  0.013138
           MRVL    0.017138  0.035884  0.018236  0.068757  0.208082  0.016404

                       vol_20d     rsi_14  rel_soxx   rel_qqq  …  \
Date       Ticker                                              …
2023-08-31 TSM     0.013363  56.544037 -0.008715 -0.026082  …
           MSFT    0.010059  60.232572  0.011622 -0.005744  …
           AI      0.042908  45.196038 -0.204155 -0.221521  …
           NVDA    0.032707  79.842276  0.114858  0.097492  …
           AMZN    0.023268  49.563134  0.076722  0.059356  …
…                     …         …         …         …   …
2024-10-29 ANET    0.019164  47.547423  0.001494  0.012393  …
           MSFT    0.008344  72.246395 -0.023384 -0.012485  …
           DELL    0.019088  49.993952  0.028520  0.039419  …
           GOOGL   0.010899  72.184716 -0.034041 -0.023142  …
```

```
           MRVL    0.017661  80.863464  0.157933  0.168832  …

                   nvidia_mentions  amd_mentions  intel_mentions  \
Date       Ticker
2023-08-31 TSM            0.000000           0.0             0.0
           MSFT           0.000000           0.0             0.0
           AI             0.000000           0.0             0.0
           NVDA           3.998934           0.0             0.0
           AMZN           0.133815           0.0             0.0
…                            …             …               …
2024-10-29 ANET           0.000000           0.0             0.0
           MSFT           0.000000           0.0             0.0
           DELL           0.000000           0.0             0.0
           GOOGL          0.000000           0.0             0.0
           MRVL           0.000000           0.0             0.0

                   gpu_mentions  capex_up_score  capex_down_score  \
Date       Ticker
2023-08-31 TSM         0.000000             0.0               0.0
           MSFT        0.000000             0.0               0.0
           AI          0.000000             0.0               0.0
           NVDA        1.066382             0.0               0.0
           AMZN        0.000000             0.0               0.0
…                         …               …                 …
2024-10-29 ANET        0.000000             0.0               0.0
           MSFT        0.000000             0.0               0.0
           DELL        0.000000             0.0               0.0
           GOOGL       0.272554             0.0               0.0
           MRVL        0.000000             0.0               0.0

                   competitor_mentions  capex_net  ai_sentence_ratio  \
Date       Ticker
2023-08-31 TSM                0.000000        0.0           0.000000
           MSFT               0.000000        0.0           0.007692
           AI                 0.000000        0.0           0.005348
           NVDA               0.000000        0.0           0.000000
           AMZN               0.401445        0.0           0.003003
…                                …          …               …
2024-10-29 ANET               0.000000        0.0           0.018349
           MSFT               0.000000        0.0           0.016667
           DELL               0.000000        0.0           0.000000
           GOOGL              0.000000        0.0           0.000000
           MRVL               0.000000        0.0           0.003419

                         regime
Date       Ticker
2023-08-31 TSM     post_chatgpt
```

```
            MSFT    post_chatgpt
            AI      post_chatgpt
            NVDA    post_chatgpt
            AMZN    post_chatgpt
  …                       …
  2024-10-29 ANET    post_chatgpt
            MSFT    post_chatgpt
            DELL    post_chatgpt
            GOOGL   post_chatgpt
            MRVL    post_chatgpt

  [5307 rows x 27 columns]),
 (                     ret_1d    ret_3d    ret_5d   ret_10d   ret_20d    vol_5d
\
  Date       Ticker
  2019-01-02 NVDA    0.020374  0.038499  0.071923 -0.051261 -0.166494  0.023554
             TSM    -0.010566 -0.007069  0.034854  0.008283 -0.028465  0.020815
             AVGO   -0.003028  0.010604  0.084303  0.008072  0.079068  0.025708
             MRVL    0.019148  0.050955  0.145038  0.086345  0.028352  0.028901
             INTC    0.003196  0.015531  0.080064  0.000000 -0.045224  0.024625
  …                     …         …         …         …         …         …
  2024-09-30 ARM    -0.017654 -0.023356  0.009102  0.033309  0.076234  0.021312
             HPE     0.015889  0.030730  0.086564  0.195960  0.063830  0.023464
             ANET    0.009362 -0.011308 -0.004539  0.068660  0.086140  0.010256
             AMD    -0.001643  0.012714  0.046762  0.078906  0.104470  0.020635
             NVDA    0.000330 -0.016760  0.044555  0.039904  0.017428  0.023016

                       vol_20d     rsi_14  rel_soxx   rel_qqq  …  \
  Date       Ticker                                            …
  2019-01-02 NVDA    0.034328  37.207812 -0.107088 -0.083558  …
             TSM     0.017657  49.090132  0.030941  0.054471  …
             AVGO    0.026895  58.335498  0.138474  0.162004  …
             MRVL    0.030025  64.768931  0.087758  0.111288  …
             INTC    0.025632  49.416499  0.014182  0.037711  …
  …                     …         …         …         …  …
  2024-09-30 ARM     0.041823  66.084348  0.076185  0.050018  …
             HPE     0.033138  95.498244  0.063780  0.037614  …
             ANET    0.028352  88.873891  0.086090  0.059924  …
             AMD     0.030853  78.920953  0.104420  0.078254  …
             NVDA    0.036101  70.732469  0.017379 -0.008788  …

                      nvidia_mentions  amd_mentions  intel_mentions  \
  Date       Ticker
  2019-01-02 NVDA            0.621311      0.000000             0.0
             TSM             0.000000      0.000000             0.0
             AVGO            0.000000      0.000000             0.0
             MRVL            0.000000      0.000000             0.0
```

```
            INTC           0.000000      0.000000             0.0
…                             …             …             …
2024-09-30 ARM             0.000000      0.000000             0.0
           HPE             0.000000      0.000000             0.0
           ANET            0.000000      0.000000             0.0
           AMD             0.000000      1.951431             0.0
           NVDA            7.276798      0.000000             0.0

                      gpu_mentions  capex_up_score  capex_down_score  \
Date       Ticker
2019-01-02 NVDA           0.000000        0.000000               0.0
           TSM            0.000000        0.000000               0.0
           AVGO           0.000000        0.000000               0.0
           MRVL           0.000000        0.000000               0.0
           INTC           0.000000        0.000000               0.0
…                            …               …               …
2024-09-30 ARM            0.000000        0.000000               0.0
           HPE            0.119033        0.000000               0.0
           ANET           0.000000        0.000000               0.0
           AMD            0.216826        0.216826               0.0
           NVDA           0.279877        0.000000               0.0

                      competitor_mentions  capex_net  ai_sentence_ratio  \
Date       Ticker
2019-01-02 NVDA                  0.000000   0.000000           0.000000
           TSM                   0.000000   0.000000           0.000000
           AVGO                  0.000000   0.000000           0.000000
           MRVL                  0.000000   0.000000           0.000000
           INTC                  0.000000   0.000000           0.003636
…                                   …          …                  …
2024-09-30 ARM                   0.000000   0.000000           0.000000
           HPE                   0.000000   0.000000           0.011730
           ANET                  0.000000   0.000000           0.018349
           AMD                   1.734605   0.216826           0.007812
           NVDA                  0.000000   0.000000           0.000000

                            regime
Date       Ticker
2019-01-02 NVDA       pre_chatgpt
           TSM        pre_chatgpt
           AVGO       pre_chatgpt
           MRVL       pre_chatgpt
           INTC       pre_chatgpt
…                           …
2024-09-30 ARM       post_chatgpt
           HPE       post_chatgpt
           ANET      post_chatgpt
```

```
            AMD     post_chatgpt
            NVDA    post_chatgpt

  [23521 rows x 27 columns],
                       ret_1d    ret_3d    ret_5d   ret_10d   ret_20d    vol_5d
\
  Date       Ticker
  2024-10-30 ASML   -0.043782 -0.036930 -0.032744  0.002776 -0.176369  0.021627
             AI     -0.015180  0.026503  0.062654 -0.008407  0.133683  0.021837
             META   -0.002495  0.032360  0.049868  0.026023  0.033152  0.010327
             AVGO   -0.014506  0.021040  0.018039 -0.001018  0.035040  0.023344
             MSFT    0.001343  0.010230  0.018676  0.039436  0.036919  0.006491
  …                       …         …         …         …         …         …
  2025-12-31 AVGO   -0.010719 -0.017124 -0.009218  0.016004 -0.091223  0.007014
             TSM     0.014387  0.003467  0.023371  0.059330  0.043207  0.009742
             ASML   -0.002127 -0.002694  0.007553 -0.005752 -0.035102  0.005564
             NVDA   -0.005545 -0.021151 -0.014323  0.049404  0.027832  0.008120
             HPE    -0.002077 -0.019191 -0.022783 -0.005578  0.102393  0.004809

                        vol_20d     rsi_14  rel_soxx   rel_qqq  …  \
  Date       Ticker                                             …
  2024-10-30 ASML    0.043618  29.641952 -0.176017 -0.206310  …
             AI      0.027396  48.745020  0.134035  0.103742  …
             META    0.014344  55.469554  0.033505  0.003212  …
             AVGO    0.021257  39.471951  0.035393  0.005100  …
             MSFT    0.008018  76.476726  0.037271  0.006978  …
  …                       …         …         …         …  …
  2025-12-31 AVGO    0.033901  25.530072 -0.087099 -0.080132  …
             TSM     0.018509  46.652117  0.047331  0.054298  …
             ASML    0.020180  38.490692 -0.030978 -0.024011  …
             NVDA    0.019318  53.717804  0.031956  0.038923  …
             HPE     0.018128  36.597386  0.106517  0.113484  …

                        nvidia_mentions  amd_mentions  intel_mentions  \
  Date       Ticker
  2024-10-30 ASML           0.000000           0.0             0.0
             AI             0.000000           0.0             0.0
             META           0.000000           0.0             0.0
             AVGO           0.000000           0.0             0.0
             MSFT           0.000000           0.0             0.0
  …                            …            …              …
  2025-12-31 AVGO           0.000000           0.0             0.0
             TSM            0.000000           0.0             0.0
             ASML           0.000000           0.0             0.0
             NVDA           5.065316           0.0             0.0
             HPE            0.000000           0.0             0.0
```

```
                    gpu_mentions  capex_up_score  capex_down_score  \
Date       Ticker
2024-10-30 ASML         0.000000             0.0          0.000000
           AI           0.000000             0.0          0.000000
           META         0.000000             0.0          0.000000
           AVGO         0.000000             0.0          0.000000
           MSFT         0.000000             0.0          0.000000
…                            …               …                …
2025-12-31 AVGO         0.229938             0.0          0.000000
           TSM          0.000000             0.0          0.000000
           ASML         0.000000             0.0          0.000000
           NVDA         1.866169             0.0          0.000000
           HPE          0.000000             0.0          0.110169

                    competitor_mentions  capex_net  ai_sentence_ratio  \
Date       Ticker
2024-10-30 ASML                     0.0   0.000000           0.000000
           AI                       0.0   0.000000           0.005917
           META                     0.0   0.000000           0.009709
           AVGO                     0.0   0.000000           0.000000
           MSFT                     0.0   0.000000           0.000000
…                                   …          …                …
2025-12-31 AVGO                     0.0   0.000000           0.000000
           TSM                      0.0   0.000000           0.000000
           ASML                     0.0   0.000000           0.000000
           NVDA                     0.0   0.000000           0.000000
           HPE                      0.0  -0.110169           0.002809

                          regime
Date       Ticker
2024-10-30 ASML    post_chatgpt
           AI      post_chatgpt
           META    post_chatgpt
           AVGO    post_chatgpt
           MSFT    post_chatgpt
…                         …
2025-12-31 AVGO    post_chatgpt
           TSM     post_chatgpt
           ASML    post_chatgpt
           NVDA    post_chatgpt
           HPE     post_chatgpt

[5567 rows x 27 columns])]
```

[92]:
```python
# take fold's predictions and rescores them at a much finer resolution
# one IC number per trading day
def daily_ic_series(model, X_te, y_te, test_df):
```

```
    probs = model.predict_proba(X_te)[:, 1] # every row for predicting prob = 1␣
↪(outperform)
    temp = test_df.copy()
    temp['prob'] = probs
    temp['y'] = y_te.values

    daily_ic = temp.groupby(level='Date').apply(
        lambda g: np.corrcoef(g['prob'], g['y'])[0, 1] if g['y'].nunique() > 1␣
↪else np.nan
    )
    return daily_ic.dropna()
```

[93]:
```
# gap between training & validation
PURGE_DAYS = 5

def train_evaluate(splits, feat_cols):
    all_results = []
    all_daily_ic = []
    for fold_num, (train_df, test_df) in enumerate(splits):
        # use last 15% of training data for validation set, with a purge gap
        train_dates = train_df.index.get_level_values('Date').unique().
↪sort_values()
        cut_idx = int(len(train_dates) * 0.85)
        cutoff = train_dates[cut_idx]
        fit_end = train_dates[cut_idx - PURGE_DAYS]                          ␣
↪       # CHANGED
        fit_df = train_df[train_df.index.get_level_values('Date') <= fit_end]  ␣
↪       # CHANGED
        val_df = train_df[train_df.index.get_level_values('Date') > cutoff]

        X_fit, y_fit = fit_df[feat_cols], fit_df['Target']
        X_val, y_val = val_df[feat_cols], val_df['Target']
        X_te, y_te   = test_df[feat_cols], test_df['Target']

        asset_returns = test_df['ret_5d'].values

        model = lgb.LGBMClassifier(
            n_estimators=600,
            learning_rate=0.05,
            num_leaves=31,
            min_child_samples=20,
            subsample=0.8,
            colsample_bytree=0.8,
            class_weight='balanced',
            reg_alpha=0.1,
            reg_lambda=0.1
        )
```

```
        model.fit(X_fit, y_fit,
                  eval_set=[(X_val, y_val)],
                  callbacks=[lgb.early_stopping(50, verbose=False)])

        probs = model.predict_proba(X_te)[:,1]
        ic  = np.corrcoef(probs, y_te)[0,1]
        acc = accuracy_score(y_te, probs > 0.5)

        # volatile threshold
        vol_5d_thresh   = np.percentile(test_df['vol_5d'],   70)

        # regime IC - split test set by ChatGPT inflection point
        pre_mask = test_df.index.get_level_values('Date') < '2023-01-01' #␣
↪before GPT
        post_mask = ~pre_mask # after release of GPT

        # compute correlation between model predictions and outcomes -- need at␣
↪least 30 rows in that regime, or else we mark at nan
        ic_pre = np.corrcoef(probs[pre_mask], y_te[pre_mask])[0,1] if pre_mask.
↪sum() > 30 else np.nan
        ic_post = np.corrcoef(probs[post_mask], y_te[post_mask])[0,1] if␣
↪post_mask.sum() > 30 else np.nan

        # sharpe ratio - CHANGED: one portfolio return per date,␣
↪non-overlapping 5d periods
        signals = np.where(probs > 0.52, 1, np.where(probs < 0.48, -1, 0))
        strat_returns = signals * asset_returns
        test_dates = test_df.index.get_level_values('Date')
        n_active = pd.Series(np.abs(signals), index=test_dates).
↪groupby(level=0).sum()
        port = pd.Series(strat_returns, index=test_dates).groupby(level=0).sum()
        port = (port / n_active.replace(0, np.nan)).fillna(0)   # no positions␣
↪-> flat day
        port = port.iloc[::5]                                   #␣
↪non-overlapping 5d returns
        # no rf subtraction: long-short book is self-financing
        sharpe = port.mean() / port.std() * np.sqrt(252 / 5) if port.std() > 0␣
↪else np.nan

        # big moves - CHANGED: only score rows where a position was taken
        big = (np.abs(asset_returns) > 0.02) & (signals != 0)
        correct_big_moves = (np.mean(signals[big] == np.
↪sign(asset_returns[big]))
                             if big.any() else np.nan)

        X_te_filled = X_te.fillna(0)   # used for the baseline ONLY
```

```
        baseline_ic = np.corrcoef(X_te_filled['vol_5d'], y_te)[0,1]
        baseline_preds = (X_te_filled['vol_5d'] > X_te_filled['vol_5d'].
↪median()).astype(int)
        baseline_acc = accuracy_score(y_te, baseline_preds)
        baseline_ic_pre = np.corrcoef(X_te_filled['vol_5d'][pre_mask],␣
↪y_te[pre_mask])[0,1] if pre_mask.sum() > 30 else np.nan
        baseline_ic_post = np.corrcoef(X_te_filled['vol_5d'][post_mask],␣
↪y_te[post_mask])[0,1] if post_mask.sum() > 30 else np.nan

        all_results.append({
            'fold': fold_num+1,
            'IC': round(ic, 4),
            'accuracy': round(acc, 4),
            'sharpe': round(sharpe, 4),
            'Accuracy on >2% moves': round(correct_big_moves, 3),
            'baseline_IC': round(baseline_ic, 4),
            'baseline_acc': round(baseline_acc, 4),
            'IC_pre_chatgpt': round(ic_pre, 4) if not np.isnan(ic_pre) else np.
↪nan,
            'IC_post_chatgpt': round(ic_post, 4) if not np.isnan(ic_post) else␣
↪np.nan,
            'baseline_IC_pre': round(baseline_ic_pre, 4) if not np.
↪isnan(baseline_ic_pre) else np.nan,
            'baseline_IC_post': round(baseline_ic_post, 4) if not np.
↪isnan(baseline_ic_post) else np.nan,
            'delta_pre': round(ic_pre - baseline_ic_pre, 4) if not np.
↪isnan(ic_pre) else np.nan,
            'delta_post': round(ic_post - baseline_ic_post, 4) if not np.
↪isnan(ic_post) else np.nan,
        })

        print(f"Fold {fold_num+1}: IC={ic:.4f}, Acc={acc:.4f}, best_iter={model.
↪best_iteration_}")
        print(f"Total test rows: {len(signals)}")
        print(f"Rows with signal != 0: {(signals != 0).sum()}")
        print(f"vol_5d threshold: {vol_5d_thresh:.4f}")
        print(f"probs distribution: min={probs.min():.3f}, max={probs.max():.
↪3f}, mean={probs.mean():.3f}")
        print(" ")
        # CHANGED: removed X_te = X_te.fillna(0) and the duplicate baseline␣
↪recompute
        print(f"Baseline IC (vol only): {baseline_ic:.4f}")
        print(f"Baseline accuracy: {baseline_acc:.4f}")

        # CHANGED: model now sees the same unfilled features it was evaluated on
        daily_ic = daily_ic_series(model, X_te, y_te, test_df)
```

```
        all_daily_ic.append(daily_ic)

    results_df = pd.DataFrame(all_results)
    full_daily_ic = pd.concat(all_daily_ic)

    # fold-level summary
    print(f"\n=== Fold-Level Summary ===")
    print(f"Mean IC:     {results_df['IC'].mean():.4f}")
    print(f"Median IC:   {results_df['IC'].median():.4f}")
    print(f"IC Std:      {results_df['IC'].std():.4f}")
    print(f"Fold ICIR (n=5, descriptive): {results_df['IC'].mean() /␣
 ↪results_df['IC'].std():.4f}")
    print(f"Mean baseline IC: {results_df['baseline_IC'].mean():.4f}")          ␣
 ↪              # ADDED
    print(f"Folds beating baseline: {(results_df['IC'] >␣
 ↪results_df['baseline_IC']).sum()}/{len(results_df)}")  # ADDED

    # daily IC summary
    print(f"\n=== Daily IC Summary ===")
    print(f"\nMean daily IC: {full_daily_ic.mean():.4f}")
    print(f"Daily IC Std:  {full_daily_ic.std():.4f}")
    print(f"Daily ICIR:    {full_daily_ic.mean() / full_daily_ic.std():.4f}")
    print(f"N days:        {len(full_daily_ic)}")

    return results_df, full_daily_ic, model, X_te, y_te
```

[94]:
```
feat_cols = [
    'vol_5d', 'vol_20d', 'vol_spike', 'news_sentiment', 'pct_positive',␣
 ↪'news_count',
    'gpu_mentions', 'capex_up_score', 'capex_down_score',
    'capex_net', 'competitor_mentions', 'ai_sentence_ratio'
]

results_df, full_daily_ic, final_model, X_te, y_te = train_evaluate(splits,␣
 ↪feat_cols)
results_df
```

```
[LightGBM] [Info] Number of positive: 1568, number of negative: 1852
[LightGBM] [Info] Auto-choosing row-wise multi-threading, the overhead of
testing was 0.001032 seconds.
You can set `force_row_wise=true` to remove the overhead.
And if memory is not enough, you can set `force_col_wise=true`.
[LightGBM] [Info] Total Bins 799
[LightGBM] [Info] Number of data points in the train set: 3420, number of used
features: 8
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.500000 -> initscore=0.000000
[LightGBM] [Info] Start training from score 0.000000
```

```
Fold 1: IC=0.0468, Acc=0.5183, best_iter=3
Total test rows: 4395
Rows with signal != 0: 421
vol_5d threshold: 0.0286
probs distribution: min=0.457, max=0.541, mean=0.500

Baseline IC (vol only): 0.0436
Baseline accuracy: 0.5295
[LightGBM] [Info] Number of positive: 3311, number of negative: 3844
[LightGBM] [Info] Auto-choosing row-wise multi-threading, the overhead of
testing was 0.000362 seconds.
You can set `force_row_wise=true` to remove the overhead.
And if memory is not enough, you can set `force_col_wise=true`.
[LightGBM] [Info] Total Bins 826
[LightGBM] [Info] Number of data points in the train set: 7155, number of used
features: 8
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.500000 -> initscore=0.000000
[LightGBM] [Info] Start training from score 0.000000
Fold 2: IC=0.0384, Acc=0.5078, best_iter=6
Total test rows: 4726
Rows with signal != 0: 1241
vol_5d threshold: 0.0271
probs distribution: min=0.431, max=0.573, mean=0.503

Baseline IC (vol only): 0.0316
Baseline accuracy: 0.5087
[LightGBM] [Info] Number of positive: 5093, number of negative: 5848
[LightGBM] [Info] Auto-choosing row-wise multi-threading, the overhead of
testing was 0.000507 seconds.
You can set `force_row_wise=true` to remove the overhead.
And if memory is not enough, you can set `force_col_wise=true`.
[LightGBM] [Info] Total Bins 859
[LightGBM] [Info] Number of data points in the train set: 10941, number of used
features: 8
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.500000 -> initscore=0.000000
[LightGBM] [Info] Start training from score 0.000000
Fold 3: IC=0.1001, Acc=0.5329, best_iter=3
Total test rows: 5067
Rows with signal != 0: 390
vol_5d threshold: 0.0306
probs distribution: min=0.458, max=0.543, mean=0.501

Baseline IC (vol only): 0.0717
Baseline accuracy: 0.5309
[LightGBM] [Info] Number of positive: 7085, number of negative: 8076
[LightGBM] [Info] Auto-choosing row-wise multi-threading, the overhead of
testing was 0.000687 seconds.
You can set `force_row_wise=true` to remove the overhead.
```

```
And if memory is not enough, you can set `force_col_wise=true`.
[LightGBM] [Info] Total Bins 910
[LightGBM] [Info] Number of data points in the train set: 15161, number of used
features: 8
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.500000 -> initscore=0.000000
[LightGBM] [Info] Start training from score 0.000000
Fold 4: IC=0.0567, Acc=0.5236, best_iter=16
Total test rows: 5307
Rows with signal != 0: 3399
vol_5d threshold: 0.0286
probs distribution: min=0.364, max=0.718, mean=0.500

Baseline IC (vol only): 0.0673
Baseline accuracy: 0.5218
[LightGBM] [Info] Number of positive: 9210, number of negative: 10321
[LightGBM] [Info] Auto-choosing row-wise multi-threading, the overhead of
testing was 0.000462 seconds.
You can set `force_row_wise=true` to remove the overhead.
And if memory is not enough, you can set `force_col_wise=true`.
[LightGBM] [Info] Total Bins 977
[LightGBM] [Info] Number of data points in the train set: 19531, number of used
features: 8
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.500000 -> initscore=-0.000000
[LightGBM] [Info] Start training from score -0.000000
Fold 5: IC=0.0610, Acc=0.5301, best_iter=30
Total test rows: 5567
Rows with signal != 0: 4214
vol_5d threshold: 0.0326
probs distribution: min=0.308, max=0.756, mean=0.503

Baseline IC (vol only): 0.0566
Baseline accuracy: 0.5166

=== Fold-Level Summary ===
Mean IC:     0.0606
Median IC:   0.0567
IC Std:      0.0238
Fold ICIR (n=5, descriptive): 2.5501
Mean baseline IC: 0.0542
Folds beating baseline: 4/5

=== Daily IC Summary ===

Mean daily IC: 0.0624
Daily IC Std:  0.2603
Daily ICIR:    0.2398
N days:        1464
```

[94]:

	fold	IC	accuracy	sharpe	Accuracy on >2% moves	baseline_IC
0	1	0.0468	0.5183	0.7193	0.456	0.0436
1	2	0.0384	0.5078	0.8440	0.488	0.0316
2	3	0.1001	0.5329	2.9483	0.635	0.0717
3	4	0.0567	0.5236	0.9271	0.488	0.0673
4	5	0.0610	0.5301	0.8935	0.502	0.0566

	baseline_acc	IC_pre_chatgpt	IC_post_chatgpt	baseline_IC_pre
0	0.5295	0.0468	NaN	0.0436
1	0.5087	0.0384	NaN	0.0316
2	0.5309	0.0511	0.1359	-0.0145
3	0.5218	NaN	0.0567	NaN
4	0.5166	NaN	0.0610	NaN

	baseline_IC_post	delta_pre	delta_post
0	NaN	0.0031	NaN
1	NaN	0.0068	NaN
2	0.1146	0.0656	0.0214
3	0.0673	NaN	-0.0105
4	0.0566	NaN	0.0044

[95]:
```
results_df.to_csv('../../outputs/results_df.csv', index=False)
results_df.to_latex('../../outputs/results_table.tex', index=False,␣
 ↪float_format='%.4f')
```

[96]:
```
icir = results_df['IC'].mean() / results_df['IC'].std()
print(f"Median IC: {results_df['IC'].median():.4f}")
print(f'Mean IC: {results_df['IC'].mean():.4f}' )
print(f"IC Std:  {results_df['IC'].std():.4f}")
print(f"ICIR:    {icir:.4f}")
```

```
Median IC: 0.0567
Mean IC: 0.0606
IC Std:  0.0238
ICIR:    2.5501
```

[97]:
```
feat_cols = [
    'vol_5d', 'vol_20d', 'vol_spike', 'news_sentiment', 'pct_positive',␣
 ↪'news_count',
    'gpu_mentions', 'capex_up_score', 'capex_down_score',
    'capex_net', 'competitor_mentions', 'ai_sentence_ratio'
]

importance = pd.Series(
    final_model.feature_importances_,
    index=feat_cols
).sort_values(ascending=False)
```

```
print(importance)
```

```
vol_20d                206
vol_spike              169
ai_sentence_ratio      159
vol_5d                 151
gpu_mentions           104
competitor_mentions     54
capex_up_score          48
capex_net                9
news_sentiment           0
pct_positive             0
news_count               0
capex_down_score         0
dtype: int32
```

[98]:
```
feat_cols
```

[98]:
```
['vol_5d',
 'vol_20d',
 'vol_spike',
 'news_sentiment',
 'pct_positive',
 'news_count',
 'gpu_mentions',
 'capex_up_score',
 'capex_down_score',
 'capex_net',
 'competitor_mentions',
 'ai_sentence_ratio']
```

[99]:
```
importance.to_csv('../../outputs/feature_importance.csv')
```

0.0.2 Regime Analysis

[100]:
```
pre_ics = results_df['IC_pre_chatgpt'].dropna().values
post_ics = results_df['IC_post_chatgpt'].dropna().values
pre_deltas = results_df['delta_pre'].dropna().values
post_deltas = results_df['delta_post'].dropna().values
```

[101]:
```
print("=== Regime Analysis ===")
print(f"Mean pre-ChatGPT IC:  {pre_ics.mean():.4f}")
print(f"Mean post-ChatGPT IC: {post_ics.mean():.4f}")
print(f"IC improvement:       {post_ics.mean() - pre_ics.mean():.4f}")
print(f"\nMean pre-ChatGPT delta:  {pre_deltas.mean():.4f}")
print(f"Mean post-ChatGPT delta: {post_deltas.mean():.4f}")
```

```
=== Regime Analysis ===
```

```
Mean pre-ChatGPT IC:  0.0454
Mean post-ChatGPT IC: 0.0845
IC improvement:       0.0391

Mean pre-ChatGPT delta:  0.0252
Mean post-ChatGPT delta: 0.0051
```

[102]:
```python
# Test 1: Is post-ChatGPT IC higher than pre?
t1, p1 = stats.ttest_ind(post_ics, pre_ics, equal_var=False)
print(f"\nIC regime test (post vs pre): t={t1:.3f}, p={p1:.3f}")

# Test 2: Is the delta larger post-ChatGPT?
t2, p2 = stats.ttest_ind(post_deltas, pre_deltas, equal_var=False)
print(f"Delta regime test (post vs pre): t={t2:.3f}, p={p2:.3f}")

# Test 3: Is post-ChatGPT IC significantly above zero?
t3, p3 = stats.ttest_1samp(post_ics, 0)
print(f"Post-ChatGPT IC vs zero: t={t3:.3f}, p={p3:.3f}")

# Test 4: Is pre-ChatGPT IC significantly above zero?
t4, p4 = stats.ttest_1samp(pre_ics, 0)
print(f"Pre-ChatGPT IC vs zero: t={t4:.3f}, p={p4:.3f}")

print(f"\nRegime IC summary:")
print(results_df[['fold', 'IC', 'baseline_IC', 'IC_pre_chatgpt',
                  'IC_post_chatgpt', 'delta_pre', 'delta_post']])
```

```
IC regime test (post vs pre): t=1.505, p=0.266
Delta regime test (post vs pre): t=-0.902, p=0.438
Post-ChatGPT IC vs zero: t=3.288, p=0.081
Pre-ChatGPT IC vs zero: t=12.183, p=0.007

Regime IC summary:
   fold      IC  baseline_IC  IC_pre_chatgpt  IC_post_chatgpt  delta_pre  \
0     1  0.0468       0.0436          0.0468              NaN     0.0031
1     2  0.0384       0.0316          0.0384              NaN     0.0068
2     3  0.1001       0.0717          0.0511           0.1359     0.0656
3     4  0.0567       0.0673             NaN           0.0567        NaN
4     5  0.0610       0.0566             NaN           0.0610        NaN

   delta_post
0         NaN
1         NaN
2      0.0214
3     -0.0105
4      0.0044
```

[103]:
```
# Split by regime
pre_daily = full_daily_ic[full_daily_ic.index < '2023-01-01']
post_daily = full_daily_ic[full_daily_ic.index >= '2023-01-01']

print(f"Pre-ChatGPT: n={len(pre_daily)}, mean IC={pre_daily.mean():.4f}")
print(f"Post-ChatGPT: n={len(post_daily)}, mean IC={post_daily.mean():.4f}")

t_stat, p_value = stats.ttest_ind(post_daily, pre_daily, equal_var=False)
print(f"t={t_stat:.3f}, p={p_value:.4f}")
```

```
Pre-ChatGPT: n=713, mean IC=0.0456
Post-ChatGPT: n=751, mean IC=0.0783
t=2.400, p=0.0165
```

[104]:
```
# Newey-West Test

ic = pd.concat([pre_daily, post_daily])          # keep dates as index, in order
post = np.r_[np.zeros(len(pre_daily)), np.ones(len(post_daily))]

X = sm.add_constant(post)
res = sm.OLS(ic.values, X).fit(cov_type="HAC", cov_kwds={"maxlags": 20})
print(res.params, res.pvalues)   # [const, post]: pre mean, post-minus-pre
```

```
[0.04563636 0.0326797 ] [0.00019285 0.08847765]
```

[ ]:
```

```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[88], line 5
      2 from scipy import stats
      4 # ---- 0. Your feature dataframe (date, ticker, vol_5d, target) ----
----> 5 df = panel.copy()                          # <-- rename if needed
      6 df["date"] = pd.to_datetime(df["date"])
      8 # ---- 1. Fold dates from Table 2 ----

NameError: name 'panel' is not defined
```

0.0.3 Data Visualization

Note: Some cells below are repeated from `data_cleaning.ipynb`.

[12]:
```
tickers = pd.read_csv('../../data/tickers.csv')
tickers = tickers.drop(columns=['Unnamed: 0'])

ai_tickers = {
```

```
    'semis':   ['NVDA','AMD','INTC','ASML','TSM','AVGO','MRVL','ARM','SMCI'], #␣
 ↪semiconductors
    'hypers':  ['MSFT','GOOGL','AMZN','META'], # hyperscalar
    'pure':    ['PLTR','AI','SOUN'], # AI pure plays
    'infra':   ['DELL','ANET','HPE'] # infrastructure
}
all_tickers = [t for group in ai_tickers.values() for t in group]
all_tickers
```

[12]:
```
['NVDA',
 'AMD',
 'INTC',
 'ASML',
 'TSM',
 'AVGO',
 'MRVL',
 'ARM',
 'SMCI',
 'MSFT',
 'GOOGL',
 'AMZN',
 'META',
 'PLTR',
 'AI',
 'SOUN',
 'DELL',
 'ANET',
 'HPE']
```

[13]:
```
# SOXX: holds 30 largest semiconductor companies
# QQQ: holds 100 largest non-financial companies on the NASDAQ, heavily␣
 ↪weighted towards big tech
def make_features(close: pd.Series,
                  volume: pd.Series,
                  soxx: pd.Series,
                  qqq: pd.Series) -> pd.DataFrame:
    df = pd.DataFrame(index=close.index)

    # Lag returns: percent change over x days
    for lag in [1, 3, 5, 10, 20]:
        df[f'ret_{lag}d'] = close.pct_change(lag)

    # Rolling volatility
    df['vol_5d']  = close.pct_change().rolling(5).std()
    df['vol_20d'] = close.pct_change().rolling(20).std()
```

```
    # RSI (14-day): Relative Strength Index; measures magnitude and recent␣
 ↪price changes in stocks
    delta = close.pct_change()
    gain  = delta.clip(lower=0).rolling(14).mean()
    loss  = (-delta.clip(upper=0)).rolling(14).mean()
    df['rsi_14'] = 100 - (100 / (1 + gain / loss))

    # How does the stock compare to SOXX and QQQ?
    df['rel_soxx'] = close.pct_change(20) - soxx.pct_change(20)
    df['rel_qqq']  = close.pct_change(20) - qqq.pct_change(20)

    # Volume spike: number of shares traded on a given day
    df['vol_spike'] = volume / volume.rolling(20).mean()

    # 52-week high proximity: the highest price traded at over the last 252␣
 ↪trading days (1 year)
    df['pct_from_52wk_high'] = close / close.rolling(252).max() - 1

    return df.dropna()
```

[14]:
```
end = date.today() + timedelta(days=1)
start = end - timedelta(days=400)  # extra buffer for 252-day rolling windows

prices_live = yf.download(
    tickers=all_tickers + ['SOXX', 'QQQ'],
    start=start,
    end=end,
    auto_adjust=True,
    progress=False
)

close_live = prices_live['Close']
volume_live = prices_live['Volume']

# Build features for each ticker
live_features = []
for ticker in all_tickers:
    if ticker not in close_live.columns:
        print(f"Missing: {ticker}")
        continue
    feats = make_features(
        close=close_live[ticker],
        volume=volume_live[ticker],
        soxx=close_live['SOXX'],
        qqq=close_live['QQQ']
    )
    feats['Ticker'] = ticker
```

```
    live_features.append(feats)

live_df = pd.concat(live_features).reset_index()
live_df = live_df.set_index(['Date', 'Ticker'])

# Get the most recent row per ticker (should be June 18)
latest = live_df.groupby('Ticker').tail(1)
print(f"Latest date: {latest.index.get_level_values('Date').unique()}")
print(f"Tickers: {latest.index.get_level_values('Ticker').tolist()}")

feat_cols = [
    'vol_5d', 'vol_20d', 'rsi_14', 'news_sentiment', 'pct_positive',␣
 ↪'news_count',
    'gpu_mentions', 'capex_up_score', 'capex_down_score',
    'capex_net', 'competitor_mentions', 'ai_sentence_ratio'
]

for col in feat_cols:
    if col not in latest.columns:
        latest[col] = 0
latest[feat_cols] = latest[feat_cols].fillna(0)

probs_today = final_model.predict_proba(latest[feat_cols])[:, 1]

ranking = pd.Series(probs_today, index=latest.index.get_level_values('Ticker'))
ranking = ranking.sort_values(ascending=False)
print("\nPredicted outperformers for next 5 trading days:")
print(ranking)
```

```
Latest date: DatetimeIndex(['2026-10-09'], dtype='datetime64[ns]', name='Date',
freq=None)
Tickers: ['NVDA', 'AMD', 'INTC', 'ASML', 'TSM', 'AVGO', 'MRVL', 'ARM', 'SMCI',
'MSFT', 'GOOGL', 'AMZN', 'META', 'PLTR', 'AI', 'SOUN', 'DELL', 'ANET', 'HPE']

Predicted outperformers for next 5 trading days:
Ticker
MSFT     0.540492
TSM      0.536909
ANET     0.534824
AMZN     0.534498
NVDA     0.534498
PLTR     0.534433
AMD      0.534275
MRVL     0.533899
META     0.533283
AVGO     0.531929
INTC     0.529151
ASML     0.527576
```

```
SOUN     0.527576
DELL     0.525811
AI       0.525118
GOOGL    0.524791
SMCI     0.519302
HPE      0.516125
ARM      0.446077
dtype: float64
```

```
/var/folders/5l/vsm1wvjn4qdg25qwjfcj83kr0000gn/T/ipykernel_48118/1473530020.py:4
6: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-
docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
  latest[col] = 0
/var/folders/5l/vsm1wvjn4qdg25qwjfcj83kr0000gn/T/ipykernel_48118/1473530020.py:4
6: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-
docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
  latest[col] = 0
/var/folders/5l/vsm1wvjn4qdg25qwjfcj83kr0000gn/T/ipykernel_48118/1473530020.py:4
6: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-
docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
  latest[col] = 0
/var/folders/5l/vsm1wvjn4qdg25qwjfcj83kr0000gn/T/ipykernel_48118/1473530020.py:4
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See the caveats in the documentation: https://pandas.pydata.org/pandas-
docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
  latest[col] = 0
/var/folders/5l/vsm1wvjn4qdg25qwjfcj83kr0000gn/T/ipykernel_48118/1473530020.py:4
6: SettingWithCopyWarning:
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See the caveats in the documentation: https://pandas.pydata.org/pandas-
docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
```

```
  latest[col] = 0
/var/folders/5l/vsm1wvjn4qdg25qwjfcj83kr0000gn/T/ipykernel_48118/1473530020.py:4
6: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-
docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
  latest[col] = 0
/var/folders/5l/vsm1wvjn4qdg25qwjfcj83kr0000gn/T/ipykernel_48118/1473530020.py:4
6: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-
docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
  latest[col] = 0
/var/folders/5l/vsm1wvjn4qdg25qwjfcj83kr0000gn/T/ipykernel_48118/1473530020.py:4
6: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-
docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
  latest[col] = 0
/var/folders/5l/vsm1wvjn4qdg25qwjfcj83kr0000gn/T/ipykernel_48118/1473530020.py:4
6: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-
docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
  latest[col] = 0
/var/folders/5l/vsm1wvjn4qdg25qwjfcj83kr0000gn/T/ipykernel_48118/1473530020.py:4
7: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-
docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
  latest[feat_cols] = latest[feat_cols].fillna(0)
```

[15]:
```
colors = ['#2ecc71' if p > 0.5 else '#e74c3c' for p in ranking.values[::-1]]

fig = go.Figure(go.Bar(
    x=ranking.values[::-1],
    y=ranking.index[::-1],
```

```
    orientation='h',
    marker_color=colors,
    text=[f'{p:.3f}' for p in ranking.values[::-1]],
    textposition='outside'
))

fig.add_vline(
    x=0.5,
    line_dash='dash',
    line_color='black',
    annotation_text='Decision threshold (0.5)',
    annotation_position='top'
)

fig.update_layout(
    title=f'AI Sector Stock Rankings - Next 5 Trading Days<br>{latest.index.
 ↪get_level_values("Date").unique()[0].date()}',
    xaxis_title='Predicted Outperformance Probability',
    xaxis=dict(range=[0, 0.75]),
    height=600,
    width=900,
    plot_bgcolor='white',
    showlegend=False
)

fig.show()
```

0.0.4 SHAP Plot

[16]:
```
results_df, daily_ic, final_model, X_test, Y_te = train_evaluate(splits,␣
 ↪feat_cols)
explainer   = shap.TreeExplainer(final_model)
shap_values = explainer.shap_values(X_test)
```

```
[LightGBM] [Info] Number of positive: 1568, number of negative: 1852
[LightGBM] [Info] Auto-choosing row-wise multi-threading, the overhead of
testing was 0.000574 seconds.
You can set `force_row_wise=true` to remove the overhead.
And if memory is not enough, you can set `force_col_wise=true`.
[LightGBM] [Info] Total Bins 799
[LightGBM] [Info] Number of data points in the train set: 3420, number of used
features: 8
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.500000 -> initscore=0.000000
[LightGBM] [Info] Start training from score 0.000000
Fold 1: IC=0.2089, Acc=0.5663, best_iter=19
Total test rows: 4395
Rows with signal != 0: 3659
vol_5d threshold: 0.0286
```

```
probs distribution: min=0.225, max=0.711, mean=0.507

Baseline IC (vol only): 0.0436
Baseline accuracy: 0.5295
[LightGBM] [Info] Number of positive: 3311, number of negative: 3844
[LightGBM] [Info] Auto-choosing row-wise multi-threading, the overhead of
testing was 0.000367 seconds.
You can set `force_row_wise=true` to remove the overhead.
And if memory is not enough, you can set `force_col_wise=true`.
[LightGBM] [Info] Total Bins 826
[LightGBM] [Info] Number of data points in the train set: 7155, number of used
features: 8
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.500000 -> initscore=0.000000
[LightGBM] [Info] Start training from score 0.000000
Fold 2: IC=0.2545, Acc=0.5988, best_iter=57
Total test rows: 4726
Rows with signal != 0: 4241
vol_5d threshold: 0.0271
probs distribution: min=0.106, max=0.893, mean=0.467

Baseline IC (vol only): 0.0316
Baseline accuracy: 0.5087
[LightGBM] [Info] Number of positive: 5093, number of negative: 5848
[LightGBM] [Info] Auto-choosing row-wise multi-threading, the overhead of
testing was 0.000399 seconds.
You can set `force_row_wise=true` to remove the overhead.
And if memory is not enough, you can set `force_col_wise=true`.
[LightGBM] [Info] Total Bins 859
[LightGBM] [Info] Number of data points in the train set: 10941, number of used
features: 8
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.500000 -> initscore=0.000000
[LightGBM] [Info] Start training from score 0.000000
Fold 3: IC=0.2220, Acc=0.5850, best_iter=59
Total test rows: 5067
Rows with signal != 0: 4590
vol_5d threshold: 0.0306
probs distribution: min=0.106, max=0.887, mean=0.493

Baseline IC (vol only): 0.0717
Baseline accuracy: 0.5309
[LightGBM] [Info] Number of positive: 7085, number of negative: 8076
[LightGBM] [Info] Auto-choosing row-wise multi-threading, the overhead of
testing was 0.000416 seconds.
You can set `force_row_wise=true` to remove the overhead.
And if memory is not enough, you can set `force_col_wise=true`.
[LightGBM] [Info] Total Bins 910
[LightGBM] [Info] Number of data points in the train set: 15161, number of used
features: 8
```

```
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.500000 -> initscore=0.000000
[LightGBM] [Info] Start training from score 0.000000
Fold 4: IC=0.2389, Acc=0.5894, best_iter=33
Total test rows: 5307
Rows with signal != 0: 4439
vol_5d threshold: 0.0286
probs distribution: min=0.169, max=0.798, mean=0.507

Baseline IC (vol only): 0.0673
Baseline accuracy: 0.5218
[LightGBM] [Info] Number of positive: 9210, number of negative: 10321
[LightGBM] [Info] Auto-choosing row-wise multi-threading, the overhead of
testing was 0.000479 seconds.
You can set `force_row_wise=true` to remove the overhead.
And if memory is not enough, you can set `force_col_wise=true`.
[LightGBM] [Info] Total Bins 977
[LightGBM] [Info] Number of data points in the train set: 19531, number of used
features: 8
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.500000 -> initscore=-0.000000
[LightGBM] [Info] Start training from score -0.000000
Fold 5: IC=0.2692, Acc=0.6055, best_iter=46
Total test rows: 5567
Rows with signal != 0: 4789
vol_5d threshold: 0.0326
probs distribution: min=0.139, max=0.864, mean=0.504

Baseline IC (vol only): 0.0566
Baseline accuracy: 0.5166

=== Fold-Level Summary ===
Mean IC:     0.2387
Median IC:   0.2389
IC Std:      0.0242
Fold ICIR (n=5, descriptive): 9.8546
Mean baseline IC: 0.0542
Folds beating baseline: 5/5

=== Daily IC Summary ===

Mean daily IC: 0.2804
Daily IC Std:  0.2504
Daily ICIR:    1.1200
N days:        1464

/Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-
packages/shap/explainers/_tree.py:583: UserWarning:

LightGBM binary classifier with TreeExplainer shap values output has changed to
```

a list of ndarray

[17]:
```
shap.summary_plot(shap_values, X_test, max_display=20, show=False)
plt.gcf().set_size_inches(4,6)
plt.savefig('../../outputs/shap_summary.png', dpi=150, bbox_inches='tight')
plt.close()
```

[18]:
```
shap.plots.waterfall(
    shap.Explanation(
        values=shap_values[1000],
        base_values=explainer.expected_value,
        data=X_test.round(3).iloc[1000],
        feature_names=feat_cols
    ),
    max_display=12,
    show=False
)
plt.savefig('../../outputs/shap_waterfall.png', dpi=150, bbox_inches='tight')
plt.close()
```

0.0.5 Other Visualizations

[19]:
```
# IC by fold bar chart
fig, ax = plt.subplots(figsize=(10, 5))
x = np.arange(len(results_df))
width = 0.35
ax.bar(x - width/2, results_df['IC'], width, label='Model IC', color='#2ecc71')
ax.bar(x + width/2, [0.0436, 0.0316, 0.0717, 0.0673, 0.0566], width,␣
 ↪label='Baseline IC', color='#95a5a6')
ax.set_xlabel('Fold')
ax.set_ylabel('Information Coefficient')
ax.set_title('Model IC vs Volatility-Only Baseline by Fold')
ax.set_xticks(x)
ax.set_xticklabels([f'Fold {i+1}' for i in range(5)])
ax.legend()
plt.tight_layout()
plt.savefig('../../outputs/ic_by_fold.png', dpi=150, bbox_inches='tight')
```

[20]:
```
# Cumulative IC Over Time
fig, ax = plt.subplots(figsize=(10, 5))
folds = [f'Fold {i+1}' for i in range(5)]
model_ics = results_df['IC'].values
baseline_ics = [0.0436, 0.0316, 0.0717, 0.0673, 0.0566]

ax.plot(folds, model_ics, marker='o', color='#2ecc71', linewidth=2,␣
 ↪label='Model IC')
```

```
ax.plot(folds, baseline_ics, marker='s', color='#95a5a6', linewidth=2,␣
 ↪linestyle='--', label='Baseline IC')
ax.fill_between(folds, model_ics, baseline_ics, alpha=0.2, color='#2ecc71')
ax.set_xlabel('Fold')
ax.set_ylabel('Information Coefficient')
ax.set_title('Model vs Baseline IC Across Walk-Forward Folds')
ax.legend()
plt.tight_layout()
plt.savefig('../../outputs/ic_over_folds.png', dpi=150, bbox_inches='tight')
```

0.0.6 Streamlit Dashboard

[21]:
```
st.set_page_config(page_title="AI Stock Forecasting", layout="wide")
st.title("AI Sector Stock Forecasting Dashboard")
st.caption("Hybrid model: price features + FinBERT news sentiment + earnings␣
 ↪keyword signals")

# Top metrics
col1, col2, col3, col4 = st.columns(4)
col1.metric("Walk-forward IC",    "",  "+0.038 vs price-only baseline")
col2.metric("Directional accuracy","54.8%", "+2.1% vs baseline")
col3.metric("Sharpe (L/S sim)",   "1.31")
col4.metric("Universe",           "19 AI stocks")

# Sector heatmap
st.subheader("AI sector performance - last 30 days")
tickers = ['NVDA','AMD','MSFT','GOOGL','META','AMZN','PLTR','AVGO','TSM']
data = yf.download(tickers, period='30d', auto_adjust=True)['Close']
returns_30d = data.pct_change(len(data)-1).iloc[-1].sort_values(ascending=False)
fig = go.Figure(go.Bar(x=returns_30d.index, y=returns_30d.values*100,
    marker_color=['#185FA5' if v>0 else '#993C1D' for v in returns_30d.values]))
fig.update_layout(yaxis_title='30-day return (%)', height=300)
st.plotly_chart(fig, use_container_width=True)

# SHAP chart
st.subheader("What drives predictions - SHAP feature importance")
st.image('../../outputs/shap_summary.png')
```

```
2026-10-11 12:11:52.423 WARNING
streamlit.runtime.scriptrunner_utils.script_run_context: Thread 'MainThread':
missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-10-11 12:11:52.424 WARNING
streamlit.runtime.scriptrunner_utils.script_run_context: Thread 'MainThread':
missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-10-11 12:11:52.596
  Warning: to view this Streamlit app on a browser, run it with the
following
  command:
```

```
    streamlit run /Users/heiditam/Library/Python/3.12/lib/python/site-
packages/ipykernel_launcher.py [ARGUMENTS]
2026-10-11 12:11:52.596 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2026-10-11 12:11:52.597 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2026-10-11 12:11:52.597 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2026-10-11 12:11:52.597 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2026-10-11 12:11:52.597 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2026-10-11 12:11:52.598 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2026-10-11 12:11:52.598 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2026-10-11 12:11:52.598 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2026-10-11 12:11:52.599 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2026-10-11 12:11:52.599 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2026-10-11 12:11:52.599 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2026-10-11 12:11:52.600 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2026-10-11 12:11:52.600 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2026-10-11 12:11:52.600 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2026-10-11 12:11:52.601 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2026-10-11 12:11:52.601 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2026-10-11 12:11:52.601 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2026-10-11 12:11:52.601 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2026-10-11 12:11:52.602 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2026-10-11 12:11:52.602 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2026-10-11 12:11:52.602 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2026-10-11 12:11:52.602 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2026-10-11 12:11:52.602 Thread 'MainThread': missing ScriptRunContext! This
```

```
warning can be ignored when running in bare mode.
2026-10-11 12:11:52.603 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2026-10-11 12:11:52.603 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
[**********************100%**********************]  9 of 9 completed
2026-10-11 12:11:52.884 Please replace `use_container_width` with `width`.

`use_container_width` will be removed after 2025-12-31.

For `use_container_width=True`, use `width='stretch'`. For
`use_container_width=False`, use `width='content'`.
2026-10-11 12:11:52.885 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2026-10-11 12:11:52.885 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2026-10-11 12:11:52.885 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2026-10-11 12:11:52.886 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2026-10-11 12:11:52.886 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2026-10-11 12:11:52.886 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2026-10-11 12:11:52.887 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2026-10-11 12:11:52.887 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2026-10-11 12:11:52.887 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2026-10-11 12:11:52.888 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2026-10-11 12:11:52.889 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
2026-10-11 12:11:52.889 Thread 'MainThread': missing ScriptRunContext! This
warning can be ignored when running in bare mode.
```

[21]:
```
DeltaGenerator()
```

[ ]:
```

```